Ashwani Singh

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linkedin.com/in/iemashwani | github.com/iemashwani | Portfolio

Professional Summary

Full Stack Developer with 1 year of experience in **Java, Spring Boot, React, REST APIs, SQL,** and **microservices**. Experienced in **Generative AI, Gemini API, RAG, semantic search, embeddings,** and **vector search**. Strong in **DSA, OOP, DBMS,** backend development, **WCAG 2.2A,** and **AWS**.

Experience

- **ETS (Educational Testing Service)** *July 2025 – July 2026*
  Software Developer Trainee(NATS) Gurugram, India
  - Developed full-stack features for the **TOEIC Link** platform using **Java, Spring Boot, JavaScript, HTML5, CSS3, and REST APIs** while improving accessibility through **WCAG 2.2A** compliant UI components.
  - Implemented Single Sign-On (SSO) with Google and LinkedIn OAuth 2.0 and JSON Web Token (JWT) session management, increasing user onboarding by **35%**.
  - Integrated **AWS X-Ray distributed tracing** across 15+ microservices-based services to identify latency bottlenecks and reduce MTTR by **40%**.
  - Resolved **15+** production issues across frontend and backend systems, ensured cross-browser compatibility, and participated in **code reviews, CI/CD workflows,** and **Agile development** practices.

Projects

- **AI Portfolio Assistant | Spring Boot, Gemini API, RAG, Semantic Search**
  Java, Spring Boot, REST APIs, Gemini API, RAG, Semantic Search, Embeddings, Vector Store [Portfolio]
  - Built a **RAG-based AI portfolio chatbot** using **Spring Boot, Gemini API, embeddings, and semantic search** to answer questions about skills, projects, experience, and education.
  - Implemented a **persistent vector store with cosine similarity search** to retrieve relevant portfolio content before generating grounded AI responses.
  - Deployed the application using **GitHub Pages + Render** integrating REST APIs, CORS, and environment-based API key management.
- **TOEIC Link – Online English Proficiency Assessment Platform**
  Java, Spring Boot, JavaScript, REST APIs, SQL, WCAG 2.2, HTML5, CSS3 [TOEIC Link]
  - Implemented **WCAG 2.2 Level A** guidelines: keyboard navigation, focus management, semantic HTML, accessible forms, and screen reader compatibility.
  - Developed scalable **RESTful APIs** with Spring Boot and responsive UI for secure test delivery, real-time score reporting, and high-volume concurrent sessions.
  - Designed **optimized SQL schemas**, API validation, structured error handling, and logging to improve reliability and observability.
- **School Management System**
  React, Java, SpringBoot, REST API, MySQL [Live Link]
  - Developed a **role-based school management system** for 3 user roles **Admin, Teacher, and Student** implementing authentication, protected routes, role-based access control, and RESTful API integration using **React, Spring Boot, and MySQL.**
  - Built **5+ core management modules** covering student registration and verification, attendance, marks, assignments, and notices, and deployed the **responsive production application on Vercel** with client-side routing and refresh-safe navigation.

Skills

Programming Languages: Java, JavaScript, Python, C++, SQL
Frameworks & Libraries: Spring Boot, Spring Security, Microservices, React, Hibernate, JPA
AI & Generative AI: Generative AI, LLM Integration, RAG, Semantic Search, Embeddings, Vector Search
Databases: MySQL, PostgreSQL, Vector Store
Cloud & DevOps: Git, GitHub, AWS, CI/CD, Agile/Scrum
Specialized: Frontend Development, Backend Development, REST API Design, WCAG 2.2A, Unit Testing, ARIA
Testing & Tools: Unit Testing, Postman, IntelliJ IDE, VS Code

Achievements

- **1450+** contest rating with **600+** problems solved on **LeetCode**.
- **800+** rating on **Codeforces** with consistent participation in competitive programming contests.
- Published peer-reviewed research paper **"Heart Disease Prediction Using Machine Learning"** at **ICICCT-2024**.

Education

- **Bachelor of Technology – Computer Science and Engineering** *July 2020 – August 2024*
  United Institute of Technology, Prayagraj, India CGPA: 7.38/10